Status	Finished
Started	Sunday, 25 October 2026, 11:04 PM
Completed	Sunday, 25 October 2026, 11:11 PM
Duration	6 mins 55 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Sequential search
d. Binary search ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By using a hash function
d. By comparing each element sequentially

Question **3**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Hash
c. Binary ✓
d. Linear

Question **4**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search starts over from the beginning
c. The search stops immediately ✓
d. The search backtracks to find duplicate elements

Question **5**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **6**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Compare the target element with the first element in the list
c. Sort the list
d. Divide the list into two equal parts

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the list contains duplicate elements
c. When the list is sorted
d. When the search algorithm finishes

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(log n)
- c. O(1)
- d. O(n log n)

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Hashing
- b. Dynamic search
- c. Binary search
- d. Linear search ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By sorting the list first
- b. By dividing the list into halves
- c. By using a hash function
- d. By comparing each element in the list sequentially ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. At the beginning of the list
- b. In the entire list
- c. In the right sublist
- d. In the left sublist ✓

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. log n
c. n ✓
d. n/2

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It does not work with negative numbers
d. It requires the list to be sorted ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(1) ✓
c. O👎
d. O(log n)

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the middle of the list
d. In the right sublist ✓